# Marcos Canedo


**Contact Information**

Instituto de Matemática Pura e Aplicada (IMPA)
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Profesional Website


**Education**

**Ph.D., Mathematics,** Pontificia Universidad Católica de Chile, Santiago, Chile — **Sep 2026**

Dissertation Advisor: Prof. Giancarlo Urzúa

Dissertation Title: *On Surfaces with QHD singularities*

**M.Sc., Mathematics,** Pontificia Universidad Católica de Chile, Santiago, Chile — **Jan 2024**

**B.S., Mathematics,** Universidad Mayor de San Andrés, La Paz, Bolivia — **June 2021**

Degree conferred with honor.

**Employment**

**Postdoctoral Researcher,** IMPA, Rio de Janeiro, Brazil — **Oct 2026–Present**

**Research Interests**

My research lie in the intersection of birational geometry and deformation theory of algebraic projective surfaces over the complex numbers. I have worked on the construction of $\mathbb{Q}$-Gorenstein smoothings of singular algebraic projective surfaces, with particular interest in degenerations that preserve the Betti numbers and their applications into problems of algebraic geometry.

**Honors and Awards**

**ANID Doctoral Fellowship,** — **2022–2026**
Agencia Nacional de Investigación y Desarrollo, Chile

**ANID Operational Research Grant,** — **2024–2026**
Agencia Nacional de Investigación y Desarrollo, Chile

**Graduated with Highest Honors,** — **2022**
Universidad Mayor de San Andrés, Bolivia

**Teaching Experience**

**Substitute Instructor**, Pontificia Universidad Católica de Chile - Chile

MAT 1610, Cálculo I — **Second Semester 2026**

**Summer Course Instructor**, Universidad Mayor de San Andrés - Bolivia

Classifying Conics, GEA - Summer School — **February 2026**

Introduction to Affine Algebraic Curves — **February 2025**

**Instructor**, Pontificia Universidad Católica de Chile - Chile

MAT 1640, Ecuaciones Diferenciales — **First Semester 2025**

MAT 1640, Ecuaciones Diferenciales — **First Semester 2024**

**Teaching Assistant**, CIMPA, V ELGA - Brazil

Surfaces with Wahl Singularities and their Many Applications — **12–23 Aug 2024**

**Teaching Assistant**, Universidad mayor de San Andrés - Bolivia

MAT 111, Álgebra I — **Second Semester 2021**

MAT 111, Álgebra I — **First Semester 2021**

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| MAT 111, Álgebra I | **Second Semester 2020** |
| MAT 111, Álgebra I | **First Semester 2020** |
| MAT 123, Geometry II | **Second Semester 2019** |
| MAT 113, Geometry I | **Second Semester 2019** |
| MAT 123, Geometry II | **First Semester 2019** |
| MAT 113, Geometry I | **First Semester 2019** |
| MAT 113, Geometry I | **Second Semester 2018** |

## Professional Activities, Outreach, and Service

**Summer Course Coorganizer**, Universidad Mayor de San Andrés - Bolivia

Geometría en las Alturas, **February 2026**
Intensive international 3 day school of algebraic geometry ∼60 participants

Introduction to affine algebraic curves, **February 2025**
Intensive 2 week course of algebraic geometry ∼30 participants

**Seminar Coorganizer**, Pontificia Universidad Católica de Chile - Chile

Seminario de Estudiantes de Postgrado en Matemáticas, **2023–2024**

**Workshop Coorganizer**, Pontificia Universidad Católica de Chile - Chile

Topology and Clay, **September 2024**

## Computer Skills

Experience in Mathematica, Python, Julia. Markup languages: LaTeX, HTML, Markdown.

**Software**—My contributions are available on my GitHub profile.

- Calculation of ℚHD discrepancies.
- Computation of complete M-resolutions of cyclic quotient singularities.

## Publications in Progress

4. **Canedo, M.**, Urzúa, G. (2026) *On ℚHD Horikawa Surfaces.*
3. **Canedo, M.**, (2026) *M-modifications of two-dimensional log-canonical singularities.*

## Submitted Publications

2. **Canedo, M.**, Urzúa, G., (2026) *Degenerations of the complex projective plane with only rational singularities*, [arXiv:2607.19348].
1. **Canedo, M.**, Urzúa, G., (2026) *Rational homology disk degenerations of elliptic surfaces*, [arXiv:2605.06668].

## Poster Presentations

2. Iberoamerican Congress on Singularities — 8 Dec 2025
Small Surfaces with WQHD singularities — Valparaíso, Chile

1. Math-Amsud School on Geometry — 4-7 Dec 2023
W-Surfaces and their MMP — El Quisco, Chile

## Contributed Talks

3. Conference: Algebraic Varieties and their Singularities, Universidad de Talca — 30 Nov 2025
Small surfaces with only WQHD singularities — Talca, Chile

2. Sesión de Geometría Algebraica, Sociedad Matemática de Chile — 25 Nov 2025
Small surfaces with only WQHD singularities — Vicuña, Chile

1. Chungnam National University — 30 Oct 2025
Classification of Small surfaces with only WQHD singularities — Daejeon, South Korea

## Invited Talks

8. Universidad de la Frontera — 18 Nov 2025
   Small surfaces with only WQHD singularities — Temuco, Chile
7. Institute for Basic Science — 13 Nov 2025
   Small surfaces with only WQHD singularities — Daejeon, South Korea
6. Seminario de Estudiantes de Postgrado - PUC — 16 Oct 2025
   ¿What is a Small Surface? — Santiago, Chile
5. Seminario de Estudiantes de Postgrado en Matemáticas - PUC — 30 Oct 2024
   Horikawa Surfaces — Santiago, Chile
4. Seminario del Grupo de Geometría Ecuador — 28 Oct 2024
   Horikawa Surfaces — ONLINE
3. Seminario de Estudiantes de Postgrado - UT — 4 Oct 2024
   Horikawa Surfaces — Talca, Chile
2. Workshop: "Toplogy and Clay" - PUC — 25-26 Sep 2024
   Development, Coorganizator — Santiago, Chile
1. Club de Matemática UC — 31 Mar 2023
   Música y Matemáticas: Acciones Musicales del Grupo Dihedral — Santiago, Chile

## Seminar Talks

**Seminar of Algebraic Geometry**, Pontificia Universidad Católica de Chile

15. Classification of Small Surfaces with only WQHD singularities — 9 Oct 2025
14. Rational Homology Disk Smoothing of $W_{p,q,r}$ part 2 — 6 Jun 2025
13. Rational Homology Disk Smoothing of $W_{p,q,r}$ part 1 — 29 May 2025
12. Deformaciones de superficies algebraicas y obstrucciones — 3 April 2025
11. Superficies singulares y no singulares — 27 March 2025
10. Fracciones continuas negativas, cadenas de Wahl, fracciones del 0 — 20 March 2025
9. La Conjetura QHD y su intento de demostración — 9 Apr 2024
8. Singularidades Terminales — 31 May 2023
7. Teorema del Cono: Singular VS No Singular-III — 24 May 2023
6. Teorema del Cono: Singular VS No Singular-II — 17 May 2023
5. Teorema del Cono: Singular VS No Singular-I — 10 May 2023
4. Geometría Birracional de Superficies — 3 May 2023
3. Geometría Birracional de Curvas y Superficies-II — 12 Apr 2023
2. Geometría Birracional de Curvas y Superficies-I — 5 Apr 2023
1. Introducción a la Geometría Birracional — 29 Mar 2023

## References

**Giancarlo A. Urzúa,** Full Professor, Pontificia Universidad Católica de Chile
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**Yulieth K. Prieto,** Assistant Professor, Pontificia Universidad Católica de Chile
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**Pedro Montero,** Assistant Professor, Universidad Técnica Federico Santamaría
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**Roberto Villaflor,** Assistant Professor, Universidad Técnica Federico Santamaría
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**Israel Morales,** Assistant Professor, Universidad de La Frontera
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